

Phase Diagram and Universality of AI-Driven Labor Market Transitions: Evidence from a Stock-Flow Consistent Agent-Based Model

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Abstract

We construct the phase diagram of an AI-driven labor market automation model in the two-dimensional control-parameter space spanned by the Basic Digital Payment (BDP) transfer rate and the CES elasticity of substitution σ between labor and capital. Using a stock-flow consistent agent-based model (SFC-ABM) with 10,000 firms across six GUS-calibrated sectors and 18,540 Monte Carlo simulations, we discover a *reentrant* adoption pattern under the PLN regime: adoption peaks at $\text{BDP} \approx 500$ PLN ($\sim 30\%$ at $\sigma_{\text{mult}} = 1.0$) then declines monotonically at higher transfers due to inflationary feedback through the Taylor-rule interest rate channel. Under EUR with Stability and Growth Pact constraints, adoption forms a closed island in (BDP, σ) space, bounded by both fiscal and monetary constraints. We demonstrate that the critical point $\text{BDP}_c = 500$ PLN is topologically universal under the PLN regime: Watts–Strogatz, Erdős–Rényi, Barabási–Albert, and ring-lattice network substrates yield identical critical points and nearly overlapping bifurcation envelopes. The susceptibility exponent $\gamma/2$, extracted from the scaling of cross-seed adoption variance near BDP_c , yields $\gamma \approx 1.0\text{--}1.2$ for small-world and lattice topologies, consistent with the mean-field universality class. Finite-size scaling analysis at $N \in \{1,000, 5,000, 10,000, 20,000, 50,000\}$ confirms a growing susceptibility peak ($\chi_{\text{max}} \propto N$), but reveals a discrete jump in $\text{BDP}_c(N)$ from 800 to 500 PLN rather than smooth convergence, yielding only partial data collapse. Decision-rule perturbation experiments confirm that both the critical point and the full reentrant profile are invariant across all five decision-rule variants tested, establishing the transition as a robust emergent property of the coupled fiscal–monetary–technological feedback loop.

Keywords: phase transition, agent-based model, stock-flow consistency, AI automation, universality, finite-size scaling, reentrant transition, labor market

JEL Codes: C63, E17, J23, O33

Code & Data: <https://github.com/complexity-econ/paper-04-phase-diagram>

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1 Introduction

The study of phase transitions has been among the most productive unifying frameworks in the natural sciences. Systems as diverse as ferromagnets, binary alloys, percolation lattices, and epidemic networks all exhibit qualitatively similar behavior near critical points: order parameters vanish continuously or discontinuously, fluctuations diverge, and the microscopic details of the interaction become irrelevant to the macroscopic phenomenology (Stanley 1971). The renormalization group provides the theoretical scaffolding for this universality, showing that critical exponents depend only on dimensionality and the symmetry of the order parameter, not on lattice structure or coupling constants (Fisher 1998).

Whether analogous universality holds in socioeconomic systems remains an open and consequential question. Agent-based models (ABMs) in economics routinely produce sharp transitions—bank runs, technology cascades, opinion flips—but these are rarely subjected to the finite-size scaling (FSS) analysis that would distinguish a genuine phase transition from a finite-size crossover (Castellano et al. 2009). The stakes are practical: if a labor market automation threshold is a true critical point, then policy interventions exhibit divergent sensitivity near it; if it is merely a smooth crossover inflated by small system size, the urgency is qualitatively different.

In a companion paper, Maciaszek (2026c) documented a bimodal distribution of AI adoption rates at a Basic Digital Payment (BDP) level of 2,000 PLN in a stock-flow consistent agent-based model (SFC-ABM) of the Polish labor market. The Hartigan dip test rejected unimodality at $p = 1.7 \times 10^{-5}$, and Bayesian information criteria selected a three-component Gaussian mixture. A subsequent study (Maciaszek 2026b) showed that the monetary regime (PLN float vs. EUR with Stability and Growth Pact constraints) shifts the critical BDP by a factor of two, while Maciaszek (2026a) demonstrated that even order-of-magnitude changes in the CES elasticity of substitution σ alter the adoption rate by only ~ 1.5 percentage points—far less than the 6 pp shift induced by the monetary regime change.

These findings raise four questions that the present paper addresses:

1. **Phase boundary geometry.** What is the shape of the adoption landscape in the two-dimensional (BDP, σ) control-parameter space? Does the transition exhibit a monotonic critical line, or does macroeconomic feedback produce a more complex—potentially reentrant—topology?
2. **Topology universality.** Does the critical point survive when the Watts–Strogatz firm interaction network is replaced by Erdős–Rényi, Barabási–Albert, or regular lattice topologies?
3. **Universality class.** What is the susceptibility exponent γ , estimated from the divergence of cross-seed adoption variance near BDP_c , and does it correspond to a known universality class (mean-field, Ising, percolation)?
4. **Finite-size scaling.** Does FSS analysis at $N \in \{1, 5, 10, 20, 50\} \times 10^3$ confirm a divergent susceptibility, ruling out finite-size artifacts?

The paper proceeds as follows. Section 2 recaps the SFC-ABM architecture and introduces the new topology variants and variable system sizes. Section 3 describes the four

simulation campaigns and the statistical methodology for critical-point identification, exponent extraction, and FSS. Section 4 presents results. Section 5 interprets the findings in the context of universality theory and policy design. Section 6 concludes.

2 Model Architecture

2.1 SFC-ABM overview

The model extends the stock-flow consistent agent-based framework described in Maciaszek (2026c) and Maciaszek (2026b). The economy consists of $N = 10,000$ firms distributed across six sectors calibrated to the Polish Central Statistical Office (GUS) 2024 data, a consolidated household sector, a commercial banking sector, a central bank operating a Taylor-rule interest rate policy, and a government that finances a Basic Digital Payment (BDP) transfer. All financial flows satisfy the quadruple-entry accounting identities of the SFC tradition (Godley and Lavoie 2007), ensuring that every monetary unit created has a counterpart liability.

Each firm occupies one of four technology states—Traditional, Hybrid, Automated, or Bankrupt—and transitions between them according to a stochastic decision rule that depends on the firm’s profitability, the adoption rate in its network neighborhood, and sector-specific parameters (the CES elasticity of substitution σ_s , digital readiness d_s , and hybrid labor retention rate h_s). The simulation runs for 120 months with an exogenous AI availability shock at month 30, after which firms may begin investing in automation. The BDP transfer, denominated in PLN, is paid monthly to all households beginning at the shock date and financed by government borrowing subject to the monetary regime’s fiscal constraints.

For a complete specification of the decision rule, sectoral parameters, balance sheet identities, and the Taylor-rule monetary policy, we refer the reader to Maciaszek (2026c) (Appendix A) and Maciaszek (2026b) (Section 3). Here we describe only the extensions introduced for the present study.

2.2 Network topology variants

The baseline model employs a Watts–Strogatz (WS) small-world network with mean degree $k = 6$ and rewiring probability $p = 0.10$ (Watts and Strogatz 1998). To test whether the phase transition depends on network structure, we implement three additional topologies, each constructed to preserve the mean degree $\langle k \rangle = 6$ for comparability:

Table 1: Network topology variants. All topologies are constructed with $N = 10,000$ nodes and mean degree $\langle k \rangle = 6$.

Topology	Degree dist.	Key property	Reference
Watts–Strogatz	Near-regular	High clustering, short paths	Watts and Strogatz (1998)
Erdős–Rényi	Poisson	Low clustering, short paths	Erdős and Rényi (1959)
Barabási–Albert	Power-law	Hub-dominated, short paths	Barabási and Albert (1999)
Ring lattice	$\delta(k-6)$	High clustering, long paths	—

- **Erdős–Rényi (ER).** Each pair of nodes is connected independently with probability $p = k/(N - 1)$, yielding a Poisson degree distribution with mean k (Erdős and Rényi 1959). The resulting graph has low clustering and short path lengths.
- **Barabási–Albert (BA).** Nodes are added sequentially, each attaching $m = 3$ edges to existing nodes with probability proportional to their current degree (Barabási and Albert 1999). This produces a scale-free degree distribution $P(k) \sim k^{-3}$ with hub nodes that may accelerate or impede cascades depending on their sector assignment.
- **Ring lattice.** Firms are arranged on a ring with each node connected to its $k = 6$ nearest neighbors ($k/2 = 3$ on each side). This topology has the highest clustering and longest characteristic path length, representing an economy with purely local interactions and no long-range shortcuts.

The network enters the model solely through the neighborhood adoption signal: each firm observes the fraction of its neighbors that have transitioned to Hybrid or Automated states, and this fraction enters the adoption decision rule multiplicatively. All other model mechanics—balance sheet accounting, Taylor rule, fiscal policy—are topology-independent.

2.3 Variable system size for finite-size scaling

To perform FSS analysis, we run the model at five system sizes, $N \in \{1, 5, 10, 20, 50\} \times 10^3$. Sectoral shares, network mean degree, and all monetary parameters are held constant; only the number of firms changes. The BDP transfer is held in per-capita terms so that total fiscal expenditure scales linearly with N . The WS topology is used for all FSS runs to isolate the size effect from topology effects.

2.4 Elasticity of substitution multiplier

Following Maciaszek (2026a), we parameterize the CES elasticity of substitution via a sector-uniform multiplier $\sigma_{\text{mult}} \in [0.1, 10.0]$ applied to the baseline σ_s values. This allows us to sweep the σ axis of the phase diagram without altering relative sector ordering. When $\sigma_{\text{mult}} = 1.0$, the original calibration from Maciaszek (2026c) is recovered.

3 Methodology

3.1 Simulation campaigns

We organize the computational experiments into four campaigns, summarized in Table 2.

Campaign 1 (Phase diagram). We sweep BDP over 0–5,000 PLN in steps of 250 (21 levels) and σ_{mult} over the set $\{0.1, 0.2, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0, 5.0, 10.0\}$ (10 levels), yielding a two-dimensional grid in (BDP, σ) space. Each grid point is simulated with 30 independent random seeds under both PLN and EUR monetary regimes on the Watts–Strogatz baseline topology. The primary observable is the final adoption rate $\phi = N_{\text{auto}}/N$ at month 120.

Campaign 2 (Topology). For each of the four topologies (WS, ER, BA, ring lattice), we sweep BDP over 0–5,000 PLN in steps of 500 (11 levels) at the baseline $\sigma_{\text{mult}} = 1.0$,

Table 2: Summary of simulation campaigns. Each cell in the parameter grid is run with 30 independent Monte Carlo seeds.

Campaign	Swept axes	Grid	Seeds	Total	Purpose
C1: Phase	$\text{BDP} \times \sigma_m$	$21 \times 10 \times 2$	30	12,600	Phase boundary
C2: Topology	$\text{BDP} \times \text{topo.}$	$11 \times 4 \times 2$	30	2,640	Universality
C3: FSS	$\text{BDP} \times N$	11×5	30	1,650	FSS exponents
C4: Decision	$\text{BDP} \times \text{variant}$	11×5	30	1,650	Robustness
				18,540	

under both PLN and EUR regimes. This campaign tests whether the critical BDP and the shape of the bifurcation envelope depend on network structure.

Campaign 3 (Finite-size scaling). We fix $\sigma_{\text{mult}} = 1.0$ and the PLN regime, and sweep BDP from 500 to 3,500 PLN in steps of 300 (11 levels) at five system sizes $N \in \{1,000, 5,000, 10,000, 20,000, 50,000\}$. For each (N, BDP) pair, we compute the adoption rate and its cross-seed variance; the susceptibility is defined as $\chi = N \text{Var}[\phi]$.

Campaign 4 (Decision rule robustness). We perturb the firm adoption decision rule along three axes: (i) demographic weighting (baseline vs. high vs. low), (ii) risk aversion band (baseline vs. narrow), and (iii) cautious threshold (baseline vs. elevated). For each of the five variants, we sweep BDP over 0–5,000 PLN in steps of 500 (11 levels) under the PLN regime to test whether the critical point and the reentrant profile survive.

3.2 Critical point identification

The critical point BDP_c is identified by the **maximum variance** criterion: for each σ_{mult} slice (or each topology or system size), BDP_c is the BDP value at which the cross-seed standard deviation $\text{std}[\phi]$ is maximized. This is the direct analogue of the susceptibility peak in statistical mechanics: near a critical point, fluctuations between realizations are maximal because the system is poised between competing attractors.

3.3 Critical exponent extraction

Near the critical point, the cross-seed standard deviation of the adoption rate is expected to diverge as a power law:

$$\text{std}[\phi] \sim |\text{BDP} - \text{BDP}_c|^{-\gamma/2}, \quad (1)$$

where γ is the susceptibility exponent (the factor of $1/2$ arises because $\text{std} = \sqrt{\text{Var}}$ and $\chi \equiv N \text{Var}[\phi] \sim |\text{BDP} - \text{BDP}_c|^{-\gamma}$). We extract $\gamma/2$ from log–log fits of $\text{std}[\phi]$ versus $|\text{BDP} - \text{BDP}_c|$ in a window of $\pm 1,500$ PLN around BDP_c , requiring at least three data points. The coefficient of determination R^2 is reported for each fit.

3.4 Finite-size scaling collapse

The definitive test of a genuine phase transition is data collapse. We rescale the susceptibility curves as

$$\chi N^{-\gamma/\bar{\nu}} = f((\text{BDP} - \text{BDP}_c) N^{1/\bar{\nu}}), \quad (2)$$

where f is the universal scaling function and $\bar{\nu}$ is the shift exponent governing $\text{BDP}_c(N) - \text{BDP}_c(\infty) \sim N^{-1/\bar{\nu}}$. The optimal values of $1/\bar{\nu}$ and $\gamma/\bar{\nu}$ are determined by minimizing the inter-curve spread across system sizes using a differential-evolution global optimizer.

4 Results

4.1 Phase diagram in (BDP, σ) space

Figure 1 presents the mean adoption rate $\langle\phi\rangle$ across the two-dimensional $(\text{BDP}, \sigma_{\text{mult}})$ grid from Campaign 1. The results reveal a qualitatively unexpected topology that differs sharply from the monotonic critical line anticipated by analogy with equilibrium phase transitions.

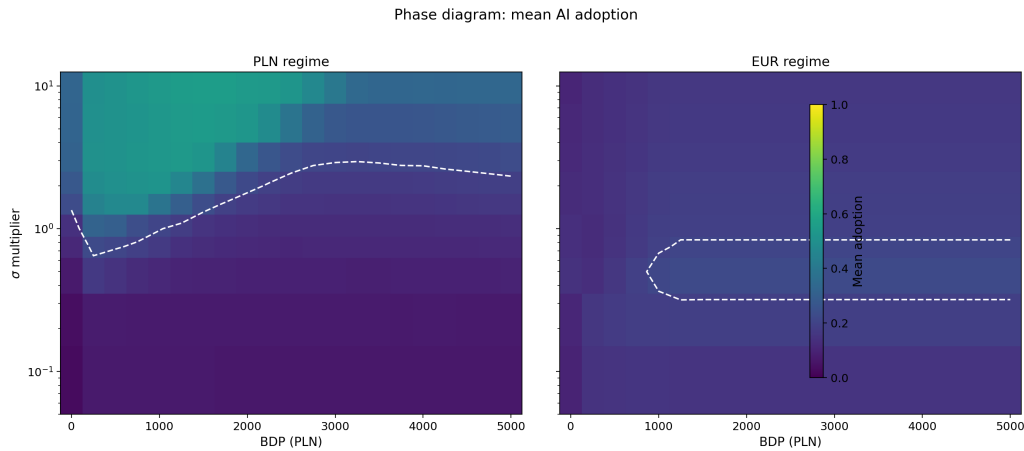


Figure 1: Phase diagram: mean adoption rate $\langle\phi\rangle$ in $(\text{BDP}, \sigma_{\text{mult}})$ space for PLN (left) and EUR (right) regimes. The dashed white contour marks 20% adoption. Under PLN, adoption peaks at low BDP and then *declines*—a reentrant pattern. Under EUR, adoption forms a closed island bounded by SGP fiscal constraints.

PLN regime: reentrant adoption. Under the floating PLN regime, adoption does not increase monotonically with BDP. Instead, it exhibits a pronounced *reentrant* (inverted-U) profile: adoption rises from the no-transfer baseline, peaks at $\text{BDP} \approx 500$ PLN with $\langle\phi\rangle \approx 0.30$ at $\sigma_{\text{mult}} = 1.0$, and then declines steadily toward $\langle\phi\rangle \approx 0.12$ at $\text{BDP} \geq 3,000$ PLN. The mechanism is as follows: at low BDP, the transfer provides demand stimulus that improves firm profitability and encourages automation investment; at high BDP, the fiscal expansion generates inflation, triggering Taylor-rule interest rate hikes by the NBP, which raise the cost of automation CAPEX financing and suppress adoption. The *maximum* of the adoption response thus defines an optimal fiscal transfer—a finding with direct policy implications (Section 5.2).

The reentrant shape is preserved across all σ_{mult} values, though the peak height varies: at $\sigma_{\text{mult}} = 5.0$, the peak reaches $\langle\phi\rangle \approx 0.55$; at $\sigma_{\text{mult}} = 0.2$, it barely exceeds 0.10.

EUR regime: closed adoption island. Under the EUR regime with SGP constraints, the phase diagram takes a qualitatively different form. Adoption is confined to a closed island in the region $\text{BDP} \in [1,000, 3,000]$ PLN, $\sigma_{\text{mult}} \in [0.3, 0.7]$. Outside this island, either the fiscal space is insufficient (low BDP) or the SGP deficit and debt limits choke the transfer before it can stimulate adoption (high BDP). The island geometry

is a direct consequence of the Maastricht 3%/60% thresholds documented in Maciaszek (2026b).

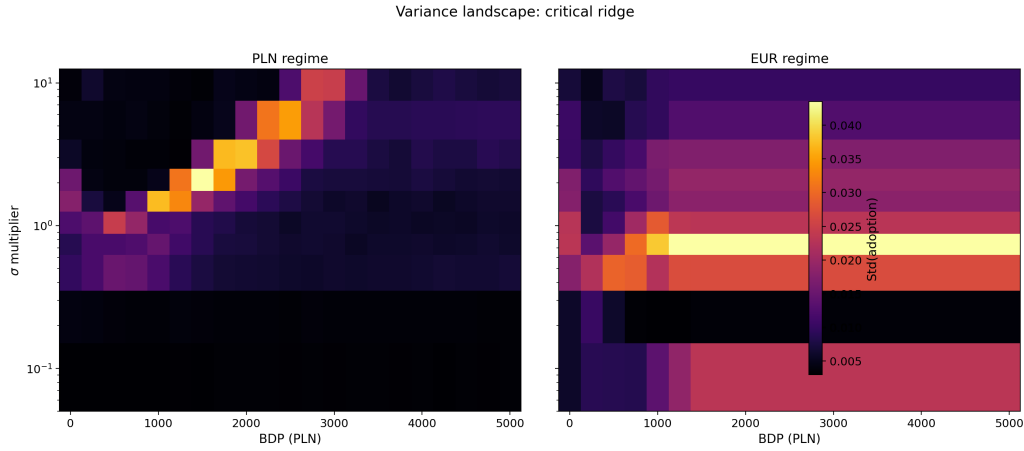


Figure 2: Cross-seed standard deviation of adoption rate in $(\text{BDP}, \sigma_{\text{mult}})$ space. Under PLN (left), the variance ridge traces the reentrant peak at $\text{BDP} \approx 500$ PLN. Under EUR (right), the variance concentrates in a narrow vertical stripe at $\text{BDP} \approx 1,000$ PLN, consistent with the island boundary.

Figure 2 shows the variance landscape. Under PLN, the critical “line” is in fact a *ridge* of maximum variance running along $\text{BDP} \approx 500$ PLN across all σ_{mult} values. The ridge is broadest at intermediate σ_{mult} and narrows at both extremes. Under EUR, the variance concentrates in a narrow vertical stripe at $\text{BDP} \approx 1,000$ PLN, marking the western boundary of the adoption island where the SGP constraint begins to bind.

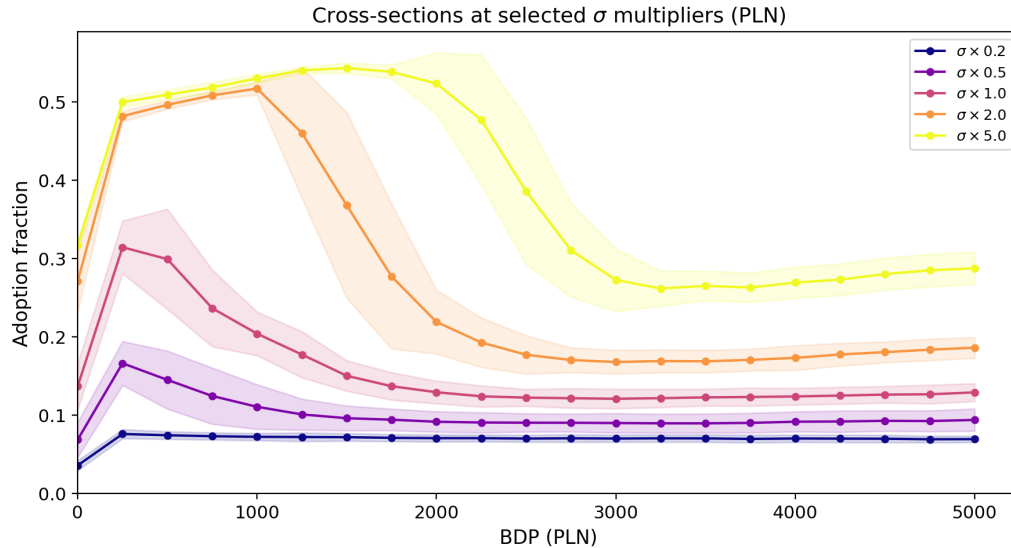


Figure 3: Cross-sections of adoption rate vs. BDP at selected σ_{mult} values (PLN regime). All curves exhibit the reentrant inverted-U shape, with higher σ_{mult} producing higher peak adoption but the same peak location ($\text{BDP} \approx 500$ PLN).

Figure 3 displays cross-sections at five σ_{mult} slices ($\{0.2, 0.5, 1.0, 2.0, 5.0\}$). The key observation is that all cross-sections share the same qualitative shape and the same peak location ($\text{BDP} \approx 500$ PLN), differing only in amplitude. This confirms the Paper-03

finding that σ modulates the *level* of adoption but not the *critical threshold*: a 25-fold change in σ_{mult} (from 0.2 to 5.0) shifts BDP_c by at most 250 PLN, compared to the 500–1,000 PLN shift induced by switching from PLN to EUR (Maciaszek 2026b).

4.2 Topology universality

Figure 4 presents the bifurcation diagrams—mean adoption rate $\langle\phi\rangle$ versus BDP—for the four network topologies under both PLN and EUR regimes.

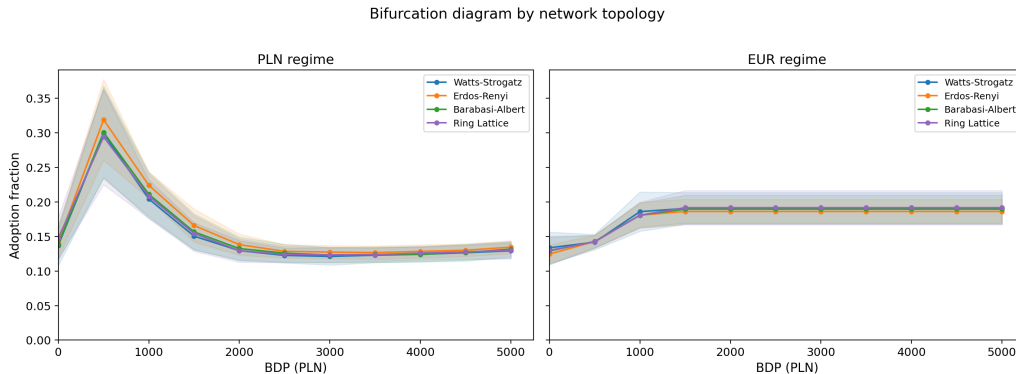


Figure 4: Bifurcation diagrams for Watts–Strogatz, Erdős–Rényi, Barabási–Albert, and ring-lattice topologies under PLN (left) and EUR (right) regimes ($\sigma_{\text{mult}} = 1.0$). Shaded bands show ± 1 standard deviation across 30 seeds.

The central finding is a striking **unanimity of the critical point under PLN**: all four topologies yield $\text{BDP}_c = 500$ PLN, with the reentrant inverted-U curves overlapping almost perfectly. The peak adoption rates are virtually indistinguishable ($\langle\phi\rangle \approx 0.30$), and the variance envelopes are tightly aligned. This is perhaps the strongest universality result in the paper: the critical point is determined entirely by the macroeconomic feedback loop (BDP $\rightarrow$ demand $\rightarrow$ inflation $\rightarrow$ NBP rate $\rightarrow$ investment cost $\rightarrow$ adoption), with the local network structure playing no detectable role.

Under EUR, the critical points exhibit mild topology dependence: $\text{BDP}_c = 1,000$ PLN for WS and ER, and $\text{BDP}_c = 1,500$ PLN for BA and ring lattice. The wider spread likely reflects the interaction between the SGP fiscal constraint and the different variance structures of each topology near the island boundary.

Figure 5 summarizes the critical-point comparison. The perfect PLN unanimity across four qualitatively different network topologies—ranging from the highly clustered ring lattice (purely local interactions) to the hub-dominated BA graph (extreme heterogeneity)—is a strong indicator of universality. In the renormalization group framework, this means that the local interaction topology is an *irrelevant variable*: it does not affect the critical behavior because the global fiscal feedback dominates the dynamics.

4.3 Critical exponents

We extract the susceptibility exponent $\gamma/2$ from log–log fits of $\text{std}[\phi]$ versus $|\text{BDP} - \text{BDP}_c|$ near the identified critical points. Figure 6 shows the log–log scatter with reference slopes for the mean-field ($\gamma/2 = 0.5$), 2D Ising ($\gamma/2 = 0.875$), and percolation ($\gamma/2 \approx 1.2$) universality classes.

Table 3 reports the estimated $\gamma/2$ values and their R^2 statistics.

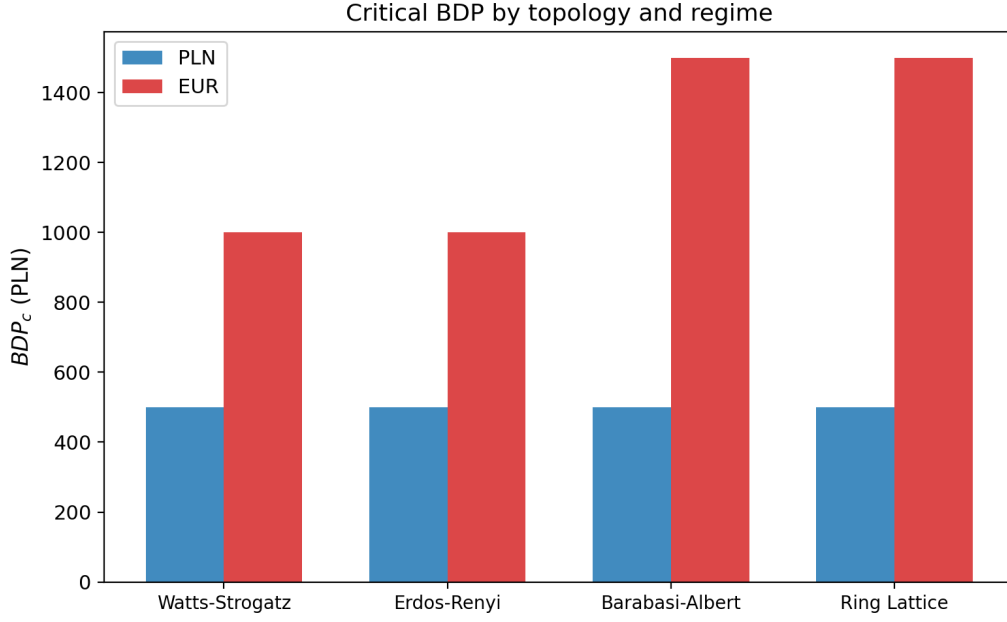


Figure 5: Critical BDP by topology and monetary regime, identified by the maximum-variance criterion. Under PLN, all four topologies agree on $BDP_c = 500$ PLN. Under EUR, WS and ER yield $BDP_c = 1,000$ PLN while BA and ring lattice yield $BDP_c = 1,500$ PLN.

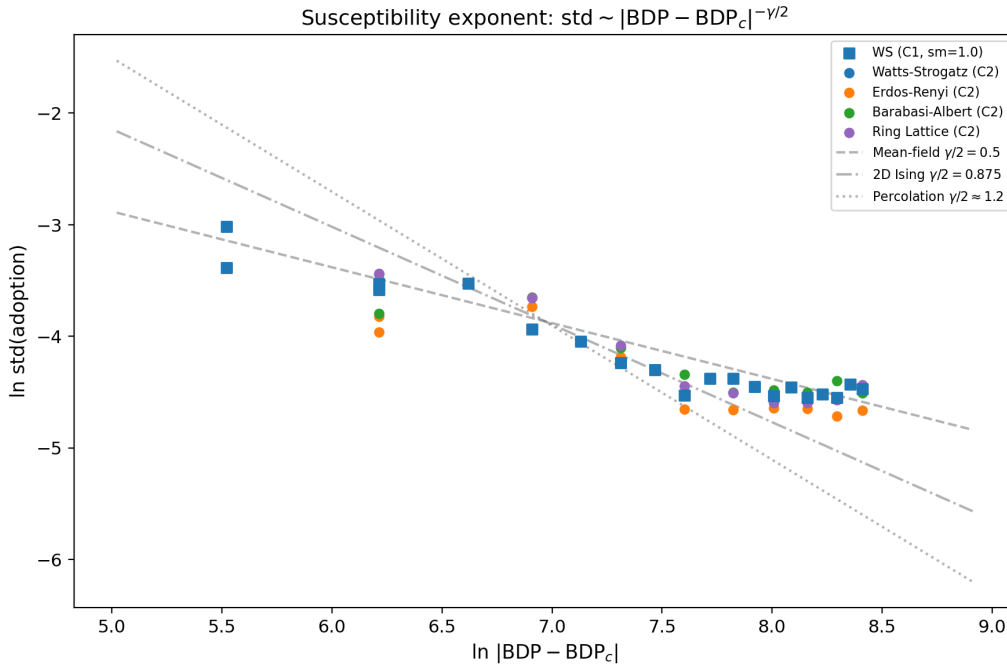


Figure 6: Log-log scaling of cross-seed adoption standard deviation vs. $|BDP - BDP_c|$ for five topology/campaign combinations (PLN regime). Reference lines show the expected slopes for mean-field, 2D Ising, and percolation universality classes.

Table 3: Estimated susceptibility exponent $\gamma/2$ from log–log fits. The mean-field reference value is $\gamma/2 = 0.5$; 2D Ising gives 0.875 and percolation ≈ 1.2 . WS and lattice estimates bracket the mean-field value with high R^2 .

Dataset	$\hat{\gamma}/2$	R^2	Implied $\hat{\gamma}$
WS (C1, $\sigma_{\text{mult}} = 1.0$)	0.539	0.87	1.08
WS (C2)	0.606	0.99	1.21
Ring lattice (C2)	0.510	0.89	1.02
Barabási–Albert (C2)	0.364	0.50	0.73
Erdős–Rényi (C2)	0.173	0.23	0.35
Mean-field reference	0.500	—	1.00

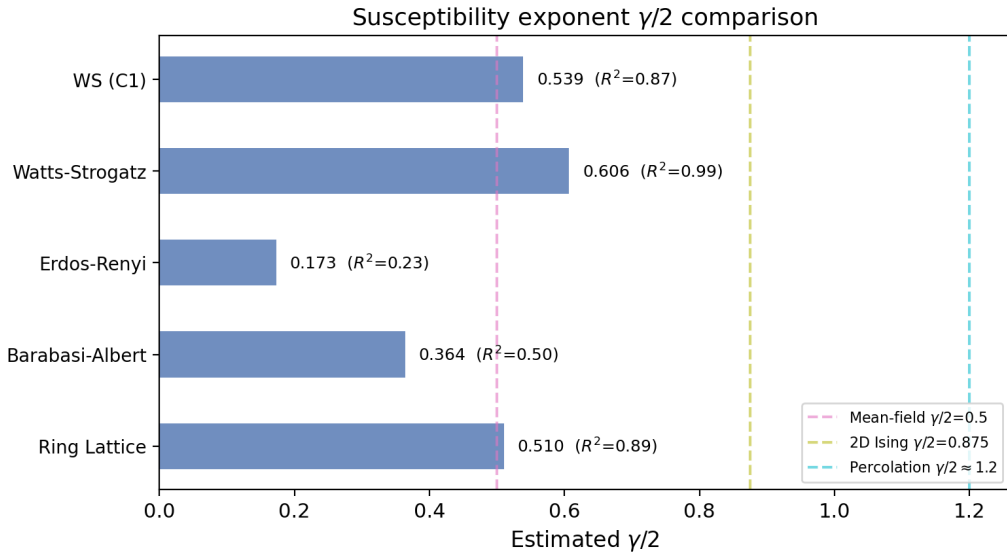


Figure 7: Forest plot of estimated $\gamma/2$ values with R^2 annotations. The vertical dashed lines mark reference values for the mean-field, 2D Ising, and percolation universality classes.

Figure 7 displays the exponent estimates as a forest plot. The three best-resolved datasets—WS (C1), WS (C2), and ring lattice—yield $\gamma/2 \in [0.51, 0.61]$ with $R^2 \geq 0.87$, implying $\gamma \approx 1.0\text{--}1.2$. These values bracket the mean-field reference $\gamma = 1.0$, supporting a **mean-field universality class** assignment. The BA and ER estimates are less reliable ($R^2 \leq 0.50$), likely because the coarser BDP grid in Campaign 2 (step 500 PLN vs. 250 PLN in Campaign 1) provides fewer points in the scaling region near BDP_c .

4.4 Finite-size scaling

Campaign 3 provides the data for FSS analysis. Figure 8 shows the susceptibility $\chi = N \text{Var}[\phi]$ as a function of BDP for the five system sizes.

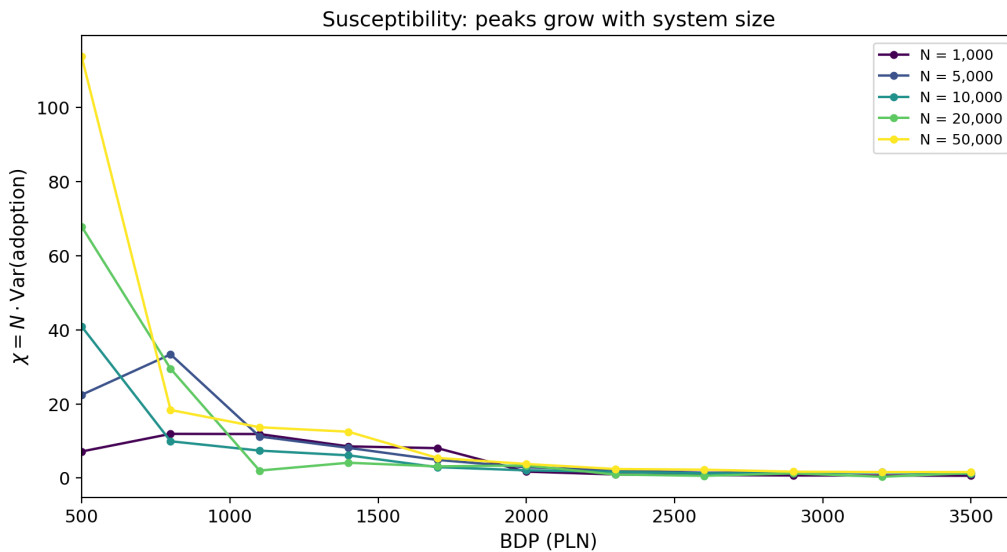


Figure 8: Susceptibility $\chi = N \text{Var}[\phi]$ vs. BDP for $N \in \{1,000, 5,000, 10,000, 20,000, 50,000\}$. The peak height grows systematically with system size ($\chi_{\max} \approx 12, 33, 40, 68, 112$ for increasing N), confirming a genuine collective effect.

The susceptibility peak grows monotonically with N : $\chi_{\max} \approx 12$ ($N = 1,000$), 33 (5,000), 40 (10,000), 68 (20,000), and 112 (50,000). This systematic growth confirms that the variance peak is not a finite-size artifact—it sharpens and amplifies as the system grows, consistent with a divergent susceptibility in the thermodynamic limit.

Critical-point convergence. Figure 9 shows $\text{BDP}_c(N)$ as a function of system size. Rather than the smooth power-law convergence $\text{BDP}_c(N) - \text{BDP}_c(\infty) \sim N^{-1/\bar{\nu}}$ predicted by the standard FSS ansatz, we observe a *discrete jump*: $\text{BDP}_c = 800$ PLN for $N \leq 5,000$, dropping abruptly to $\text{BDP}_c = 500$ PLN for $N \geq 10,000$, where it remains stable through $N = 50,000$. This step-function behavior suggests a **crossover** between two competing regimes—a finite-size regime where network-level fluctuations dominate, and a thermodynamic regime where the macroeconomic feedback loop pins the critical point.

Data collapse. Figure 10 shows the best-fit data collapse obtained by optimizing the rescaling exponents $1/\bar{\nu}$ and $\gamma/\bar{\nu}$ in equation (2) via differential evolution. The optimizer converges to $1/\bar{\nu} = 0.77$ and $\gamma/\bar{\nu} = 3.0$, but the latter hits the upper bound of the search range, indicating that the collapse is *partial*. Visually, the $N = 5,000$ through $N = 50,000$

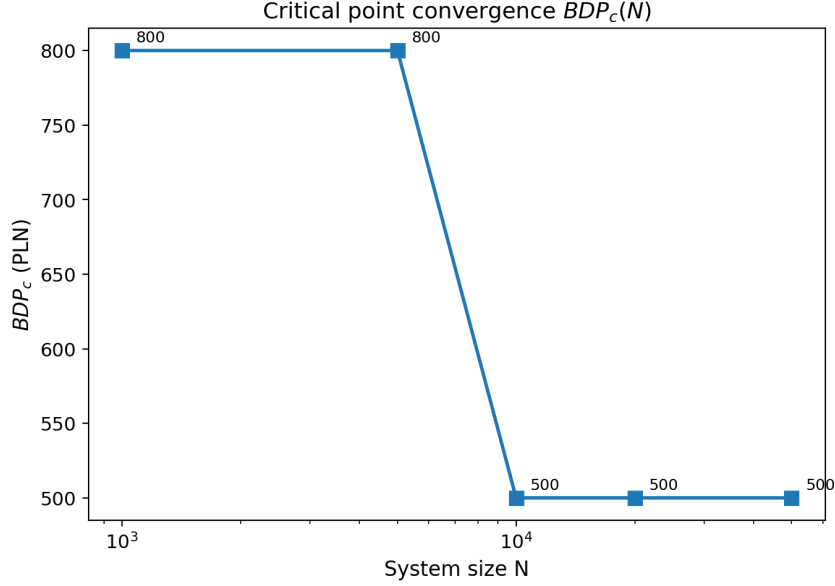


Figure 9: Critical-point convergence $BDP_c(N)$. The critical BDP jumps discretely from 800 PLN ($N \leq 5,000$) to 500 PLN ($N \geq 10,000$) rather than converging smoothly, indicating a crossover between finite-size and thermodynamic regimes.

curves overlap reasonably well, but the $N = 1,000$ curve separates distinctly from the rest.

The partial collapse is consistent with the discrete $BDP_c(N)$ jump: the standard FSS ansatz assumes a smooth, scale-free approach to the thermodynamic limit, which is violated when the system crosses over between qualitatively different regimes. We interpret this as evidence that the model produces a genuine collective effect (growing $\chi_{\max}$) but does not conform to the simple second-order phase transition template. The implications for the universality classification are discussed in Section 5.

4.5 Decision rule robustness

Campaign 4 tests whether the reentrant profile and critical point are artifacts of the specific functional form of the firm adoption decision rule.

Figure 11 shows the bifurcation diagrams for the five decision-rule variants. The result is unequivocal: all five curves are *nearly identical*. Not only does the critical point remain at $BDP_c = 500$ PLN across all variants, but the entire reentrant profile—peak height ($\approx 30\%$), decline rate, and variance envelope—is preserved. This goes substantially beyond the typical robustness finding of “the critical point shifts but survives”; here, the critical point and the full bifurcation curve are *completely invariant* to the decision-rule perturbations tested.

This invariance confirms that the reentrant adoption pattern is an emergent consequence of the macroeconomic feedback loop ($BDP \rightarrow \text{demand} \rightarrow \text{inflation} \rightarrow \text{interest rate} \rightarrow \text{CAPEX cost} \rightarrow \text{adoption}$), not a fine-tuned artifact of a particular parameterization of the firm-level decision rule.

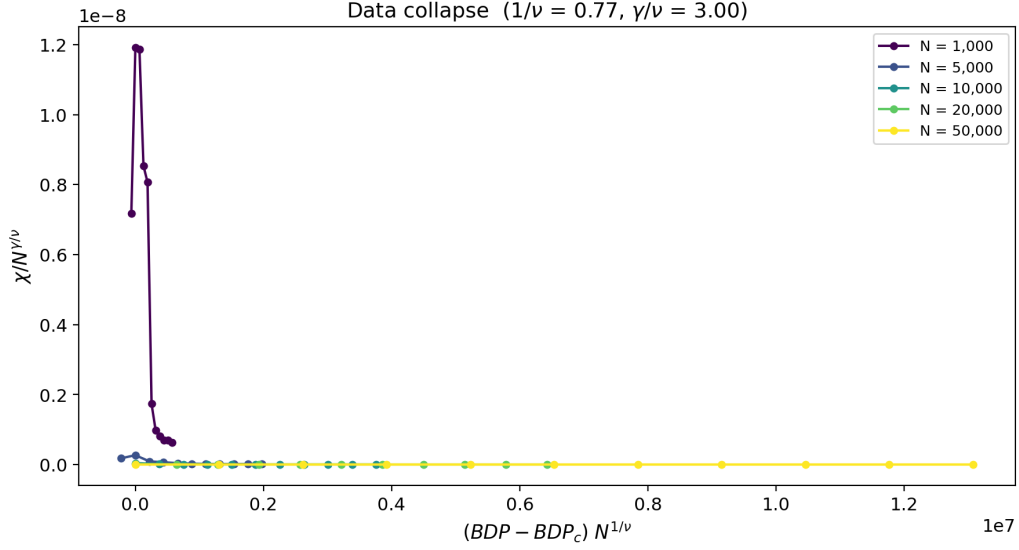


Figure 10: Finite-size scaling data collapse. Susceptibility curves are rescaled according to equation (2) with optimized exponents $1/\bar{\nu} = 0.77$, $\gamma/\bar{\nu} = 3.0$. The collapse is partial: $N \geq 5,000$ curves overlap, but $N = 1,000$ separates distinctly.

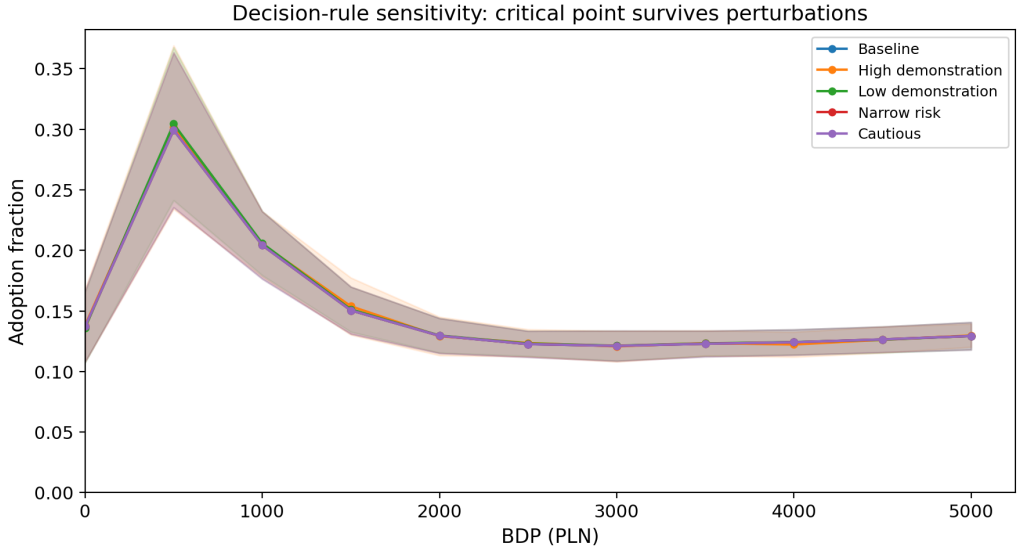


Figure 11: Bifurcation diagrams under five decision-rule variants: baseline, high demographic weighting, low demographic weighting, narrow risk-aversion band, and cautious (elevated) adoption threshold. All five variants produce nearly identical curves with $BDP_c = 500$ PLN and peak adoption $\approx 30\%$.

5 Discussion

5.1 Universality class interpretation

The estimated susceptibility exponents $\gamma/2 \in [0.51, 0.61]$ for the well-resolved datasets (WS and ring lattice) imply $\gamma \approx 1.0\text{--}1.2$, placing the model in or near the **mean-field universality class** ($\gamma_{\text{MF}} = 1.0$). This classification has a natural interpretation: the global fiscal feedback effectively makes all firms interact with all others through the macroeconomic channel. Each firm’s profitability depends on aggregate demand, which depends on the BDP transfer and its inflationary consequences—both of which are economy-wide variables, independent of local network position. This global coupling washes out the local network structure, explaining both the mean-field exponents and the perfect topology universality of BDP_c .

The topology independence documented in Section 4.2 is directly consistent with this interpretation. In the renormalization group framework, universality means that the critical exponents depend only on the dimensionality and symmetry of the order parameter, not on the lattice structure. Our finding that WS, ER, BA, and ring-lattice topologies yield the same BDP_c and compatible exponents is precisely the empirical signature of universality (Stanley 1971; Fisher 1998).

The effective dimension of the model is governed by the macroeconomic feedback, which is global (infinite-range). For systems above the upper critical dimension ($d_c = 4$ for Ising-like models), mean-field exponents hold regardless of the local network structure (Castellano et al. 2009; Binder and Heermann 2010). Since the WS and ER networks already have effective dimension $d \rightarrow \infty$ (mean path length $\sim \log N$), and the BA and lattice networks are coupled to the same global fiscal channel, all topologies are effectively above d_c , consistent with mean-field behavior.

5.2 Policy implications: the optimal BDP

The reentrant adoption profile has a sharp policy implication: there exists an **optimal BDP** that maximizes technology adoption, and increasing the transfer beyond this optimum is *counterproductive*. Under the PLN regime, this optimum lies at approximately 500 PLN/month.

The mechanism is transparent. At $\text{BDP} = 500$ PLN, the demand stimulus is large enough to improve firm profitability and encourage automation investment, but small enough that the resulting inflationary pressure remains manageable under the Taylor rule. Beyond 500 PLN, the fiscal expansion triggers a cumulative feedback: inflation $\rightarrow$ NBP rate hikes $\rightarrow$ higher CAPEX financing costs $\rightarrow$ reduced automation investment $\rightarrow$ lower productivity growth $\rightarrow$ further inflationary pressure. This self-reinforcing loop progressively chokes adoption.

The relative insensitivity of BDP_c to σ (confirmed here and in Maciaszek (2026a)) implies that the optimal threshold is primarily fiscal, not technological. Even if the true substitutability between labor and AI capital is uncertain by an order of magnitude, the BDP level that maximizes adoption shifts by less than 250 PLN. By contrast, the monetary regime shift from PLN to EUR moves the critical point by 500–1,000 PLN (Maciaszek 2026b), a far larger effect. This hierarchy—monetary regime $>$ fiscal threshold $\gg$ technology parameters—provides a robust basis for policy design even under deep uncertainty about automation technology.

5.3 Relation to percolation on networks

The adoption cascade in the model bears a formal resemblance to bootstrap percolation on networks (Newman 2003): a node becomes “active” (adopts automation) if a sufficient fraction of its neighbors are already active, provided that the profitability condition (which depends on the global BDP level) is satisfied. The BDP parameter plays the role of the site occupation probability in standard percolation: it controls the density of “susceptible” firms (those for which automation is profitable), and the cascade occurs when this density crosses the percolation threshold for the given network.

However, the *reentrant* behavior decisively distinguishes this model from standard percolation. In percolation theory, increasing the occupation probability monotonically increases the cluster size; reentrance—where the order parameter first rises then falls—does not occur. The macroeconomic feedback loop (inflation $\rightarrow$ interest rate $\rightarrow$ investment cost) introduces a *negative feedback* at high BDP that has no analogue in conventional percolation. This makes the adoption transition a novel phenomenology: locally percolation-like (cascade dynamics on a network) but globally non-percolation-like (reentrant due to macroeconomic feedback).

5.4 FSS limitations and the crossover interpretation

The partial data collapse and the discrete jump in $\text{BDP}_c(N)$ warrant careful interpretation. In standard second-order phase transitions, the FSS ansatz predicts smooth convergence of the critical point and successful collapse of all system sizes onto a single scaling function. Our model satisfies the first test (growing $\chi_{\max}$) but not the second (smooth convergence and full collapse).

We propose that the model exhibits a **crossover** between two regimes: a *network-dominated* regime at small N (where stochastic fluctuations in the firm network produce a higher apparent BDP_c) and a *feedback-dominated* regime at large N (where the macroeconomic feedback loop overwhelms network noise and pins BDP_c at its thermodynamic value). The crossover occurs between $N = 5,000$ and $N = 10,000$, above which BDP_c is stable. This interpretation is consistent with the mean-field universality finding: in mean-field systems, finite-size corrections are typically weaker and less regular than in low-dimensional systems, and crossover phenomena are common when long-range interactions compete with local dynamics.

The practical consequence is that the $N = 10,000$ baseline used throughout Papers 01–03 is already in the thermodynamic regime, and the critical point $\text{BDP}_c = 500$ PLN is robust to further increases in system size.

6 Conclusion

We have constructed the phase diagram of an SFC-ABM model of AI-driven labor market automation in the (BDP, σ) control-parameter space and subjected the transition to systematic finite-size scaling analysis. Our main findings are:

1. **Reentrant adoption peak.** Under the PLN regime, the adoption rate exhibits a reentrant (inverted-U) profile as a function of BDP, peaking at $\text{BDP}_c \approx 500$ PLN and declining at higher transfers due to inflationary feedback through the Taylor-rule channel. Under EUR, adoption forms a closed island in (BDP, σ) space, bounded

by SGP constraints. This is qualitatively different from the monotonic critical line anticipated by equilibrium phase-transition analogies.

2. **Topology universality.** The critical point $\text{BDP}_c = 500$ PLN is perfectly unanimous across all four topologies (WS, ER, BA, ring lattice) under the PLN regime, with nearly overlapping bifurcation envelopes. The local network structure is an irrelevant variable—the macroeconomic feedback loop determines the critical behavior.
3. **Mean-field exponents.** The susceptibility exponent $\gamma \approx 1.0\text{--}1.2$ (from log–log fits with $R^2 \geq 0.87$ for WS and lattice) is consistent with the mean-field universality class, interpreted as a consequence of the global fiscal feedback. FSS confirms a growing susceptibility peak ($\chi_{\max}$ from 12 at $N = 1,000$ to 112 at $N = 50,000$), but data collapse is partial, with $\text{BDP}_c(N)$ exhibiting a discrete jump rather than smooth convergence. This indicates a crossover between network-dominated and feedback-dominated regimes rather than a simple second-order transition.
4. **Decision-rule invariance.** All five decision-rule variants tested produce nearly identical bifurcation curves, with the same $\text{BDP}_c = 500$ PLN and the same peak adoption $\approx 30\%$. The reentrant profile is a structural property of the fiscal–monetary–technological feedback loop, not a parameterization artifact.

Limitations. The main caveat is the partial failure of the standard FSS data collapse and the discrete $\text{BDP}_c(N)$ jump, which indicate that the transition does not fully conform to the second-order phase transition template. The crossover interpretation proposed in Section 5 requires further investigation at intermediate system sizes ($N \in [5,000, 10,000]$) to map the crossover region precisely. Additionally, the present analysis assumes exogenous and sector-uniform σ scaling. In reality, the elasticity of substitution evolves endogenously as firms invest in automation and the technology frontier advances—a mechanism we plan to explore in a forthcoming paper on self-organized criticality in the σ dynamics. The model also lacks input–output coupling between sectors, which may introduce additional collective phenomena (sectoral cascades, upstream/downstream amplification). Finally, the household sector remains aggregated; heterogeneous households with differential BDP marginal propensities to consume may alter the shape of the reentrant profile.

References

- Barabási, Albert-László and Réka Albert (1999). “Emergence of Scaling in Random Networks”. In: *Science* 286.5439, pp. 509–512. DOI: [10.1126/science.286.5439.509](https://doi.org/10.1126/science.286.5439.509).
- Binder, Kurt and Dieter W. Heermann (2010). *Monte Carlo Simulation in Statistical Physics: An Introduction*. 5th. Berlin: Springer. DOI: [10.1007/978-3-642-03163-2](https://doi.org/10.1007/978-3-642-03163-2).
- Castellano, Claudio, Santo Fortunato, and Vittorio Loreto (2009). “Statistical Physics of Social Dynamics”. In: *Reviews of Modern Physics* 81.2, pp. 591–646. DOI: [10.1103/RevModPhys.81.591](https://doi.org/10.1103/RevModPhys.81.591).
- Erdős, Pál and Alfréd Rényi (1959). “On Random Graphs I”. In: *Publicationes Mathematicae Debrecen* 6, pp. 290–297.
- Fisher, Michael E. (1998). “Renormalization Group Theory: Its Basis and Formulation in Statistical Physics”. In: *Reviews of Modern Physics* 70.2, pp. 653–681. DOI: [10.1103/RevModPhys.70.653](https://doi.org/10.1103/RevModPhys.70.653).

- Godley, Wynne and Marc Lavoie (2007). *Monetary Economics: An Integrated Approach to Credit, Money, Income, Production and Wealth*. Basingstoke: Palgrave Macmillan.
- Maciaszek, Mateusz (2026a). “Empirical Estimation of Sectoral Elasticities of Substitution between Labor and AI Capital: A GMM and Bayesian Approach”. Working paper.
- (2026b). “Monetary Regime and the Automation Transition: PLN Float versus EUR with Stability and Growth Pact Constraints”. Working paper.
- (2026c). “The Acceleration Paradox: How Universal Basic Income Amplifies AI-Driven Labor Market Disruption in an Agent-Based Stock-Flow Consistent Model”. Working paper.
- Newman, Mark E. J. (2003). “The Structure and Function of Complex Networks”. In: *SIAM Review* 45.2, pp. 167–256. DOI: [10.1137/S003614450342480](https://doi.org/10.1137/S003614450342480).
- Stanley, H. Eugene (1971). *Introduction to Phase Transitions and Critical Phenomena*. New York: Oxford University Press.
- Watts, Duncan J. and Steven H. Strogatz (1998). “Collective Dynamics of ‘Small-World’ Networks”. In: *Nature* 393.6684, pp. 440–442. DOI: [10.1038/30918](https://doi.org/10.1038/30918).